Andrew Boldi

Solving hard science problems with machine learning

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Education

UC Berkeley *Expected May 2027*

BS in Chemical Biology & BA in Computer Science

Fall '26:	Deep Neural Networks (CS 182), Computer Vision (CS 180), Physical Chemistry (Statistical Mechanics), Molecular Biology Lab
Chemistry:	Computational Chemistry, Physical Chemistry (Quantum Mechanics), Honors Organic Chemistry I/II, Honors General Chemistry I/II
Computer Science:	ML (CS 189), AI (CS 188), Data Structures & Algorithms (CS 61B), Computer Architecture (CS 61C), Discrete Math & Probability Theory (CS 70), Structure & Interpretation of Computer Programs, Designing Information Devices & Systems
Sciences:	Macromolecular Biology, Chemical Biology (Biochemistry), Biological Inorganic Chemistry, Human Physiology, Mechanics, Electricity & Magnetism, Chemical Engineering Process Analysis, Mathematics & Statistics for Chemical Engineering

Publications

A. Boldi[†], A. K. Gupta[†], Y. Tang, M. Hamadani, K. Ameri, Y. Yeghiazarians. "Fragment-Centric Explainability of Chemical Foundation Models with Attention-Aware Layer-Wise Relevance Propagation." *Under review*, 2026. [†]Co-first authors.

Work Experience

AI Intern *South San Francisco, CA*

Soley Therapeutics: Molecular Machine Learning under direction of Dr. Ankur Gupta *May 2025 – Present*

- Full-time summer 2025 and 2026 intern; received return offer and continued projects part-time during the academic year
- **Co-first author** (manuscript under review) and **inventor** (filed patent application) on a fragment-centric explainability framework for chemical foundation models; co-led its development, applied it to Ames mutagenicity, hERG inhibition, and lipophilicity models, and demonstrated usefulness in internal lead optimization
- Leading development of explainability methods for computer vision models on cell images
- Finetuned hundreds of models on public and proprietary data to predict molecular target candidate profiles (TCPs); built an automated framework for configuration generation, finetuning, and cross-fold validation of several foundation models across 22 datasets (TDC ADMET group)
- Developing generative chemistry models for de novo molecule generation and lead optimization of therapeutics
- Implemented and benchmarked several architecture ideas for molecular property prediction foundation models; refactored the codebase of an existing foundation model
- Automated CDD (Collaborative Drug Discovery) protocol creation with Selenium + Python, completing a task projected to take 100+ manual hours in 5–6 hours
- Integrated the above projects into the product with the UI/Infra team
- Tools used: PyTorch, PyTorch Lightning, Hydra/OmegaConf, CDD Vault, AWS (EC2, S3, CodeBuild)

Research Experience

Summer Research Intern *Pasadena, CA*

Caltech: Professor Brian M. Stoltz Lab Group under direction of Dr. Scott Virgil *Jun 2023 – Aug 2023*

- Completed 16 consecutive reaction steps toward an intermediate in the total synthesis of the diterpenoid scabrolide B
- Developed software application for cataloging catalysts & ligands (StremLigand)
- Performed computational modeling (ORCA) for determining selectivity in a late-stage stereoinversion
- Used lab techniques including HPLC, LC/MS, GC/MS, LC-TOF, NMR, gCOSY 2D NMR, TLC, extraction, column chromatography, rotary evaporator, distillation, dry box

Summer Research Intern *Pasadena, CA*
Caltech: Professor Brian M. Stoltz Lab Group under direction of Dr. Scott Virgil *Jun 2022 – Aug 2022*

- Completed the 10-step synthesis of a fragment in the total synthesis of korormicin
- Completed the 8-step synthesis of a fragment in the total synthesis of ineleganolide
- Conducted computational modeling of key intermediates to identify optimal synthesis routes using ORCA
- Taught myself CAD (FreeCAD) overnight, then measured, modeled, and 3D printed a replacement gear for a broken HPLC autosampler motor the same day; refined it to an exact fit, avoiding a $250 part with a two-week lead time, and printed spares now also used in a Caltech neurobiology lab

Assistant Researcher *Remote*
Caltech: Center for Catalysis and Chemical Synthesis under direction of Dr. Scott Virgil *Jul 2020 – May 2022*

- Developed software program for remote automation of robotic autosamplers (Python, VBA, M68k assembly language)
- Reverse-engineered M68k assembly code to write more efficient functions for use in lab
- Used program to screen rare earth metal (lanthanide)-ligand catalysts for an asymmetric reduction

Research Summaries & Proposals

Computational Total Synthesis Project Proposal *Mar 2024*

Deep Reinforcement Learning for Total Synthesis

Synthetic Efforts Towards the Total Synthesis of Scabrolide B *Aug 2023*

2023 Summer Research

Synthesis of Korormicin and Ineleganolide Fragments *Aug 2022*

2022 Summer Research

Projects

Molecular Optical Rotation Prediction with Machine Learning — *Trennen* *2022*

- Self-directed project: distilled 8M+ molecules down to ∼27,000 optically active molecules using PubChem's API, RDKit, and Open Babel
- Built a T-Net (point-cloud) classifier on 3D molecular structures to predict rotation direction
- Explored computing polarizability tensors with ORCA at the B3LYP/def2-SVP level of theory (DFT)
- Tools used: TensorFlow, PubChem REST API, Open Babel, RDKit, ORCA

Scenic Navigation Algorithm with Deep Learning — *Pathfinder* *2022*

- Developed navigation algorithm (modified A*) to find scenic routes in San Mateo County
- Developed web-scraping program with Selenium to automatically extract 250K+ Google Street View images from San Mateo County
- Used a pretrained semantic segmentation model for inference on 250K+ images with Google TPUs
- Accounted for curvature of the earth with mathematical expressions for better heuristic estimation
- Processed road connections and intersections with United States Census Bureau's TIGER/Line Shapefiles
- Tools used: Python, BDD100K Dataset, Google TPUs

Pacman with Bitcoin Miner (Top Project in Berkeley Course) — *Video Demo* *Spring 2024*

- Built a Pacman game with a built-in Bitcoin miner with a project partner in 3000+ lines of Java code
- Selected as one of the top 15 projects in a class of ∼1400 students for a data structures & algorithms course from UC Berkeley (CS 61B, Spring 2024)

Math Competition Training Website — *AMC Trainer* *2021 – Present*

- Co-developed interactive website and backend with Ethan Boldi to help students prepare for math competitions (grades 8/10/12 and invitational AIME)
- Actively used by 20K+ users in 50+ countries
- Tools used: JavaScript, HTML, CSS, Cloudflare Workers, Cloudflare KV Database, Node.js

Representation of the Fibonacci and Prime Numbers as Music — *Fibprimes* *2021*

- Built a program in C to automatically write MIDI files with the pattern of Fibonacci and prime numbers

- Recognized on Y Combinator's Hacker News and earned 26 stars on GitHub
- Tools used: C, MIDI file format

Skills

Programming:	Python, Java, C/C++, Rust, Svelte, JS/CSS/HTML, M68k Assembly, Swift, LaTeX
ML & Tools:	PyTorch, PyTorch Lightning, TensorFlow, Hydra/OmegaConf, Selenium, AWS (EC2, S3, CodeBuild), Git/GitHub, Slurm, Linux
Chemistry:	Multistep organic synthesis, laboratory characterization (HPLC, LC/MS, NMR), computational modeling (ORCA, DFT), RDKit, AutoDock Vina
Languages:	English, Armenian, Spanish

Additional Experience

Physician Shadowing at Kaiser Permanente *Summer 2026*

- Shadowed Kaiser physicians and engineers in general surgery, magnetoencephalography (MEG), hospital medicine, and palliative medicine

Code Mind Berkeley: Co-founder *Nov 2023 – Jan 2024*

- Interdisciplinary research for high school students with UC Berkeley professors
- Built website with Svelte + SvelteKit + TypeScript + SCSS + GSAP
- Developed research project proposals at the intersection of machine learning, analytical chemistry, physics, biochemistry, marine science, and medicine
- Developed web-scraping pipeline for generating leads in Python + Selenium, later rewritten in Rust
- Secured $250K in client commitments (75 clients); wound down the company to focus on full-time studies

Y Combinator AI Startup School *Jun 2025*

- Selected as one of 2000 top CS undergrads, master's, and PhD candidates in AI to attend the conference, with speakers including Elon Musk, Satya Nadella, Sam Altman, Andrej Karpathy, Andrew Ng, and more
- Further invited as one of ∼20 students to have dinner with YC general partner Aaron Epstein

Private Tutor *2018 – Present*

- Teaching 7 students math and piano weekly
- Taught 25+ students in the past in subjects including linear algebra and chemistry

STEMPower: Workshop Facilitator & Program Director *Jul 2021 – Jun 2023*

- Led hands-on workshops through Zoom to promote scientific interest for grades 3–6
- Subjects taught include math, chemistry, biology, and physics

YAPA Kids: Math Tutoring Teacher & Program Director *May 2020 – Aug 2021*

- Led and taught interactive math classes for enhanced understanding of school topics
- Math Co-Program director for over 460 students from May 2021 to August 2021

Special Interests

Music *YouTube Channel*

- Multiple 1st place wins at piano competitions such as US Open Music, Armenian Allied Arts Association, Crestmont Conservatory Piano, and Satori Young Pianist (30+ recognitions)
- Performed at Carnegie Hall and Merkin Hall in New York City as a winner of the Armenian Youth Talent Competition
- Special recognition twice as Most Promising Young Talent at Armenian Allied Arts Association Competition
- Honorable mention at US International Music Competition
- Featured performer at Junior Bach Festival in Berkeley, California (and another 20+ public performances)
- Conductor of 200+ person pep band at Carlmont High School football games (Fall 2022)
- Pianist for Carlmont High School Jazz Ensemble (2019–2023)

- Studying classical piano since the age of 7
- At 12 years old, completed the final advanced level of the MTAC Certificate of Merit Program in Piano with branch honors and highest performance scores for all repertoire and perfect score on music theory exam

Rubik's Cube Puzzles

- Solved 3x3 in 9.42 seconds & have solved Rubik's cube blindfolded
- Skilled at rapidly solving 2x2, 3x3, 4x4, 5x5, 6x6, 7x7, Megaminx (12-sided), and Square-1 cubes